ESTIMATION OF THE SOCIAL COSTS OF TRAIN DELAYS IN THE NETHERLANDS

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1. INTRODUCTION

Train delays are not only frustrating for passengers, they also cost a lot of money. In 2015, some 14 million hours were lost due to delays of passenger trains in the Netherlands. The cost associated with this loss were more than € 145 million (KiM, 2016), which is a low estimate, as only the actual cost of the longer travel time due to delays had been included. The aim of this paper is to present a complete assessment of the social cost of train delays for 2016 that also includes costs associated with the variability of travel times, the costs related to the evasive reactions of passengers in response to major disruptions, and the costs due to increased discomfort when trains become more crowded as a result of disruptions. Finally, delay costs in freight transport, and additional operating costs for rail operating companies and the infrastructure manager, are part of our research (KiM, 2017).

The reliability of travel time is regarded to be one of the key performance indicators of a transport network, both for car and public transport. A lack of travel time reliability is what travellers notice: “that frustrating characteristic of the transportation system that prompts passengers to allow an hour to make a trip that normally takes 30 minutes because the actual trip time is so unpredictable”. The OECD (2010) defines reliability as “the ability of the transport system to provide the expected level of service quality upon which users have organized their activity”. The key aspect of this definition is the assumption that network users have expect a particular level of service and that reliability is a measure of the extent to which the traveller’s experience matches their expectation (Hellinga, 2011). In other words, reliability is equivalent to the predictability of travel times and, from the perspective of a traveler, associated with the statistical concept of variability.

Investments in infrastructure or other transport measures are often made because they reduce travel time, but also because they enhance reliability. In the Netherlands, transport infrastructure projects and other transport policies are appraised using cost-benefit analysis (CBA). To incorporate reliability improvements in project and policy evaluation, the Dutch Ministry of Infrastructure and the Environment includes the societal benefits of reliable and predictable travel times in the CBA). It has become common practice in the

Netherlands to monitor the social costs of congestion. Since 2004, the costs of unreliability are included in the calculations. However, for rail travel a similar monitoring does not yet exist. This paper aims to fill that gap.

Section 2 of this paper presents a framework of all relevant social costs. In section 3 some of the literature which was used to underpin this framework is described. Section 4 shows the most complete range of social costs of train delays in the Netherlands, based on the framework of section 2. Where possible, existing evidence has been used to determine the costs. Where no evidence was available, qualitative estimates have been made. The paper ends with conclusions and recommendations for future research (section 5).

2 OVERVIEW OF SOCIAL COSTS

Relevant costs of train delays can be split up between costs for passengers, costs for shippers, costs for operating companies (both passenger and freight transport), and costs for the infrastructure manager. Social costs are determined by monitoring the losses of travel time and converting these into costs by adapting a value of time. In addition to time costs, different types of expenditures may also be relevant. Table 1 shows nine components of social costs that are included in our research. Each component is followed by a short explanation.

Passengers	
1	Costs due to loss of time by train delays
2	Costs due to additional loss of time in door-to-door travel
3	Costs due to unreliability and uncertainty
4	Costs due to evasive behaviour
5	Costs due to discomfort
Shippers	
6	Costs due to delays and unreliability in rail freight transport
Operators	
7	Operating costs passenger transport companies
8	Operating costs freight transport companies
Infrastructure manager	
9	Additional costs for infrastructure management

Table 1: components of social costs for different stakeholders

Ad 1
Costs resulting from loss of time due to train delays, including cancellation of trains.

Ad 2
Costs, additional to “ad 1”, due to an extended egress travel time as a consequence of train delays or train cancellations. Most train trips have some

type of egress transport. As far as cars, bikes or walking are concerned, train delays are directly transmitted to the total door-to-door travel time. In the case of other public transport serving as an egress mode, train delays may be resolved if there is sufficient interchange time, but they may also cause further delays in the total travel time if connections are missed.

Ad 3
Costs as a result of the variability of travel times and unreliability of the services. This means uncertainty for passengers regarding their expected arrival times. Passengers are not only disadvantaged when their travel time is extended, but also when the actual travel time is unpredictable. Examples include stress, uncertainty about connections or connecting flights, missed appointments, etc. This is why people often take an earlier train in order to ensure they will arrive on time.

Ad 4
Costs as a result of (major) disruptions in train traffic that are known to the public in advance. Passengers may search for alternatives: a different route, time of departure or a different mode. Or they may refrain from travelling at all.

Ad 5
Costs related to a decreasing likelihood of finding a seat on a train that is overcrowded due to disruptions. In the perception of passengers, such trips seem to last much longer.

Ad 6
Costs as a result of longer journey times and unreliability in rail freight transport.

Ad 7
Passenger transport operators have extra costs as a result of delays and disruptions of train traffic. They include refund arrangements, expenditures for substitute transport, hotels and catering or additional staff costs. Most of these costs are transferred as income for other parties and as such they do not represent costs for society as a whole..

Ad 8
Staff costs and rolling stock costs are examples of additional costs for rail freight operating companies.

Ad 9
The infrastructure manager has extra costs in order to cope with disruptions and to redirect train traffic. An example is staff present in traffic control centres.

3. SOME LITERATURE ON THE VALUE OF TRAIN DELAYS

Having defined the nine components of the social cost of delays, we shall now look briefly at some of the relevant literature concerning the valuation of each component separately.

Component 1: Time loss due to train delays
Determining the costs of lost travel time is not complicated: it is standard practice in cost-benefit analysis to measure the amount of travel time lost by the train passengers, and then multiply that with an appropriate value-of-time. The lost travel time can be measured or estimated in various ways, while the value-of-time can be taken from the KiM (2013) report where a separate value is given for the train mode.

Component 2: Additional time loss door-to-door travel time
The valuation of this component is more difficult, and we are not aware of any evidence in the literature about how additional travel time losses in door-to-door travel due to train unreliability should be quantified and valued in a scientific manner. Although several authors mention the effect of missed public transport connections following a train delay, this topic has received little attention until now.

Component 3: Costs due to unreliability and uncertainty
Noland and Small (1995) were among the first authors describing how travel time uncertainty affects the choice of the timing of travel and hence the cost of travel. This applies to travel by car and by public transport. Travel by public transport has the additional complication that the service is a scheduled one: travellers can only depart and arrive at specific moments in time. This leads to scheduling cost (Fosgerau 2009), even for a totally reliable service, in addition to the cost of travel time itself.

If the public transport service is unreliable, departure and arrival times will deviate at random from the timetable, leading to lost time. Fosgerau and Engelson (2011) describe the value of such travel time deviations, measured by means of the standard deviation, for travellers using scheduled services. Passengers counter the effect of variability by departing earlier from home (buffer time), in order to take an earlier connection. Tseng et al. (2012) describe the consequences for cost-benefit analysis: travel times increase, scheduling cost increase, and there is also cost associated with the pure variability (measured by means of the standard deviation) of travel time itself. Lijesen (2014) describes how travellers who know about the unreliability of the train service can adjust their travel times and substantially reduce the negative impact of unreliability on scheduling costs. But the cost of the increased travel times, and the uncertainty due to the variability of travel times itself, remains.

How can we therefore quantify the cost of the variability of travel times? In the literature we have found three different types of approaches to determining the cost of the variability of travel times:

1. Determine the standard deviation of travel times, and multiply by the Value of Reliability of travel time.

2. Determine the lost travel times, and multiply these by a factor larger than one to account for the unreliability effect.
3. Determine the number of passengers departing early, and how much earlier they depart, in order to establish the average buffer time. Multiply this average buffer time by an appropriate value of buffer time.

The first approach is by far the most common one in the literature, and described for instance by Fosgerau and Karlström (2010) and Fosgerau and Engelson (2011). Here the cost of unreliability was determined by measuring the standard deviation of travel times. This is then multiplied using an appropriate Value-of-Reliability (VoR). The VoR is typically established through a large Stated Preference (SP) survey in which train passengers are asked to evaluate a series of train journeys with different combinations of expected mean travel time, variations in travel time, and fares. The KiM has conducted such an SP study, and reported the results for use in cost-benefit analysis in the Netherlands (KiM 2013). Below we present the values for passenger travel by rail mentioned in the report:

Trip Purpose	VoT	VoR	Reliability Ratio
Home-to-work	11.50	4.75	0.4
Business	19.75	22.75	1.1
Other	7.00	4.50	0.6
Average	9.25	5.50	0.6

Table 2: VoT and VoR for train trips, Euro/person/hour, market prices, price level 2010. Source: KiM (2013).

The second approach has been used by Rietveld et al. (2001). Using a Stated Preference survey among some 750 train passengers, they estimated the total cost of delayed travel time as worth 2.4 times the pure cost of travel time. Consequently, apart from the lost travel time itself, they estimated that another 1.4 times the cost of the lost travel time should be added for the uncertainty effect (standard deviation of travel time) and possible negative rescheduling effects.

The third approach is a very recent one, and reported in Li (2017). Li conducted a survey among existing train passengers to measure:

- the percentage of train passengers that actually departed earlier from home than strictly necessary, in order to counter the risk of arriving late at their destination;
- how much earlier the departure was for those passengers that did depart earlier.

Using these two simple indicators, she established the average buffer time for train journeys. By multiplying these with a value of early schedule delay the cost associated with the unreliability can be established. The order of magnitude of the value of early schedule delay early has been estimated by Fosgerau (2016), following Small (1982), as 0.5 times the Value-of-Time.

Component 4: Costs due to evasive behaviour
We are not aware of any recent literature describing methods to estimate the cost of evasive behaviour of train passengers in reaction to train delays. A pragmatic method could be to analyse the total number of train trips and travel time lost per day, and make a comparison between days without and days with important recorded train delays.

Component 5: Costs due to discomfort
In the literature there are several sources that describe how travel time multipliers have been derived to quantify the discomfort experienced by train passengers when travelling in a crowded train. Multipliers vary between 1.4 and 1.8 (Kroes et al.) and between 2.4 and 2.8 (Wardman en Whelan 2011). Here we would not just need to quantify the discomfort itself, but the *change in discomfort due to train delays*. In other words: we need to quantify the multiplier in normal conditions (trains on time) and in delay conditions (trains late), and determine the difference. The costs associated with this difference in multiplier value can then be attributed to the train delays.

Component 6: Costs due to delays and unreliability in freight travel
The KiM (2013) publication provides Values-of-Time and Values-of-Reliability for freight transport by rail, for containerised and non-containerised transport. If the standard deviation of travel time delays is known, the VoRs can be applied in exactly the same manner as described under component 3: the first method to quantify the costs of delays and unreliability for freight transport.

Components 7, 8 and 9: Additional operating costs passenger transport, additional operating costs freight transport, and additional costs infra management
We are not aware of any literature about these costs.

4 SOCIAL COSTS OF TRAIN DELAYS IN THE ‘NETHERLANDS´

In this section we show the resulting most complete range of social costs of train delays in the Netherlands, based on the framework of section 2, the literature findings of section 3, specific data provided by operating companies, and our own recent survey on passengers’ response to (possible) disruptions of train services.

Component 1: Time loss due to train delays
NS (Dutch Railways) is the dominant operating company in the Netherlands, with a market share of approximately 94% (expressed in passenger kilometres). In the Netherlands a national chip card system is in operation, with passengers checking both in and out for their trips via nearly all modes of public transport. In cooperation with ProRail, the national infrastructure manager, NS has calculated the number of so called 'passenger delay minutes'. This implies the difference between the actual journey time of all passengers compared to journey times according to passenger information systems. The effects of train cancellations are included in this method.

In 2016, the number of passenger delay minutes on the NS network was 685 million (source: Dutch Railways). This figure is expanded by some 20% due to the share of non-chipcard tickets. Based on an average Value of Time of € 10.24 for 2016 (source: KiM), the outcome is an amount of more than € 140 million in delay costs.

This outcome is completed with the share of the other four Dutch train operating companies (Arriva, Syntus, Breng and Connexxion) that account for the remaining 6% of market share. On average, unpunctuality and train cancellations are half as frequent in this segment than at NS. We estimated the additional figure for delay costs to be approximately 6/94 * 0,5 * 140 = € 4 million.

Component 2: Additional time loss door-to-door travel time
We estimated to what extent train delays are transmitted in the total door-to-door travel time. The calculation implies the following steps:

1. The number of train journeys, distinguished between arrivals in large cities and other parts of the country;
2. The share of public transport other than train, as egress mode in each of the two segments;
3. The assumption that the average additional delay (caused by the train delay) in the entire door-to-door journey is half the headway of the egress mode. We assumed 5 minutes in large cities and 15 minutes elsewhere.

The calculation resulted in an additional € 9 million in delay costs.

Component 3: Costs due to unreliability and uncertainty
We applied each of the three approaches as outlined in section 4. The first approach is to determine the standard deviation of travel times and multiply by a Value of Reliability of travel time of 0.6, as depicted in table 2. Since recent data on variation patterns of train arrivals were not available, we applied similar 2008 figures (Leijsen, 2013) and expanded them to the 2016 situation. The result is an estimated amount of € 169 million.

In the second approach we multiplied the costs of train delays as explained in components 1 and 2 by a factor 2.4. This reflects the value of uncertainty and unreliability of travel times that passengers consider frustrating, on top of longer journey times (Koopmans, 2010 and Rietveld, 2004). The result of the calculation is an amount of € 214 million.

The third approach consisted of a survey among train passengers. The main question was to what extent they depart earlier for their train trips in order to cope with possible delays and thus a late arrival - see text box. We multiplied the average buffer time with a value of 'schedule delay early' (SDE) of 0.5 times the Value of Time (Fosgerau, 2016) and concluded an estimate of € 227 million to account for coping with the risk of arriving too late.

By combining the results of the three approaches we estimate the costs of unreliability and uncertainty between € 169 and € 227 million.

Over half of train passengers occasionally include buffer time in their journeys

The KiM has conducted a survey among some 2,700 people who travel by train at least once a year. The sample is taken from the Dutch Mobility Panel, which is a fair reflection of the Dutch population.

Nearly 55% of the respondents do sometimes take an earlier train in order to ensure they do not arrive late at their destinations. On average they leave home 26 minutes earlier than strictly necessary according to the train timetable. In expanding these figures, we took the frequency of this behaviour into account. The inclusion of buffer times not only occurs in the case of work-related trips, but also private trips, which may be of such an urgent nature that passengers leave home earlier than strictly necessary, with examples being an appointment with a doctor or catching a flight.

Component 4: Costs due to evasive behaviour

Our presumption was that the number of journeys on days with large disruptions in train traffic would be significantly smaller than on other, undisturbed days. In order to test this presumption, we combined NS data on daily variations in patronage and passenger delay minutes with information on days with extensive disruptions. We were not able to establish significant differences in passenger volumes. It seems to be that most passengers just commence their intended train journey, possibly via another route or on a different time of the day, with or without a delay. These passengers form part of the situation described in component 1. Only passengers who changed to another mode may have been confronted with additional time costs. This can be concluded from additional survey questions on evasive behaviour - see text box below. These extra time costs are estimated between € 22 and 38 million. Apart from time costs, passengers sometimes may have incurred extra out of pocket costs in order to accommodate a disrupted train journey. This amount varies between € 22 and € 39 million. The total evasive costs therefore vary between € 44 and € 77 million.

In the case of disruptions known in advance, most people commence their train journeys regardless

Over 70 percent of respondents commence their train journeys even if they know in advance that the train services are disrupted. The majority of them will actually experience travel time delay. This delay, however, is already part of the calculation described in component 1. More than 11% of the respondents travelled with another mode, arriving 40 minutes late on average at their final destination. Nearly 10% of the respondents travelled at another time or refrained from travelling at all. Some 6% mentioned extra out of pocket costs following a disrupted journey.

Component 5: Costs due to discomfort

Train delays often lead to more crowded trains with less chance of finding a seat and hence greater discomfort. In the perception of passengers, such journeys seem to last longer. Using NS data on variations in the probability of finding a seat, we tested if on days with major disruptions these chances were significantly smaller than on other days. If this were the case, we can monitor the extra time effect by applying appropriate multipliers. Indeed, during days with disruptions of services the average probability of a passenger finding a seat decreases. The effect on total perceived travel time and on time costs, however, is very small from a national perspective compared to other cost components. Of course, this result may be quite different for individual train services.

Component 6: Costs due to delays and unreliability in freight travel

Only data for international freight transport were available, which accounts for 92 percent of the total rail freight transport in the Netherlands (source: CBS). In 2016, freight trains leaving the Netherlands sustained 18,683 hours of delay at the borders. This includes departure delays and delays during the journey within the Netherlands. For trains entering the Netherlands, only data on delays within the country are available: some 10,101 hours. The Value of Time for a freight train is on the average € 1,406 (prices for 2016). This includes costs for both shippers and operating companies (KiM, 2013). We calculated an amount of over € 40 million of delay costs, excluding costs of delays in other countries. Additionally, no data on variations in journey times were available, which makes it impossible to calculate unreliability costs for freight. For both these reasons the amount of costs being calculated for component 6 is an underestimate.

Component 7: Costs for passenger transport companies
Most costs for operating companies are 'transferred costs' and are therefore not to be regarded as social costs (see section 2). No data were available on additional staff costs or costs related to managing operations in case of disruptions.

Component 8: Costs for freight transport companies
Additional operating costs are included in the Value of Time analysis of component 6.

Component 9: Costs for the infrastructure manager
No data on additional costs for the infrastructure manager were available.

SUMMARY OF RESULTS
Table 3 below summarizes the results of the calculations. A part of the costs could not be included. Therefore, the total amount is an underestimate; it has been rounded off to avoid the impression of absolute accuracy.

Component		**Result**
1	Loss of time due to delays	€ 144 million
2	Door-to-door effect	€ 9 million
3	Uncertainty and unreliability	€ 169 - 227 million
4	Evasive behaviour	€ 44 - 77 million
5	Discomfort	€ 0
6	Loss of time in freight transport	€ 40 million
7	Operating costs passengers	No quantification
8	Operating costs freight	Included in 6
9	Costs infrastructure management	No quantification
Total costs in 2016, rounded off, excluding components 7 and 9		**€ 400 - 500 million**

Table 3: Summary of results. Source: KiM

5 CONCLUSIONS

We draw the following conclusions from our research:

- We have distinguished nine different components, of very different importance. The two most important components are the costs of lost travel time, and the effects of rescheduling and pure uncertainty about the departure and arrival times of trains.

- The costs of rescheduling and pure variability (with corresponding uncertainty) are higher than just the costs of the longer travel time. Valuing only the additional travel time is clearly wrong. In the case of the Netherlands, costs of travel time variability exceeds the costs of the longer travel time by a factor between 1.1 and 1.5, reflecting the importance of this indicator.

- The costs of train delays in the Netherlands are substantial: between € 400 and 500 million. Moreover, these figures are an underestimate, because some of the costs could not be quantified. This supports the policy of transport authorities of requesting that the operating companies provide a minimum standard of punctuality.

- Neither literature nor quantitative estimates were found for the additional operating costs for operators or the infrastructure manager.

- Compared with an estimated € 2.6 - 3.4 billion of congestion costs for car travel on the main network (KiM, 2017) and after correction for the smaller market share of rail, the costs of train delays in the Netherlands are relatively low.

- It would be useful to compare the results of the case of the Netherlands with similar research conducted elsewhere or with costs of road congestions. This could be a topic for future research.

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